

Problems of Vasc and Arqady

Amir Hossain Parvardi †

January 6, 2011

Edited by: Sayan Mukherjee

1. Suppose that a, b, c are positive real numbers, prove that

$$1 < \frac{a}{\sqrt{a^2 + b^2}} + \frac{b}{\sqrt{b^2 + c^2}} + \frac{c}{\sqrt{c^2 + a^2}} \leq \frac{3\sqrt{2}}{2}$$

2. If a, b, c are nonnegative real numbers, no two of which are zero, then

- $2 \left(\frac{a^3}{b+c} + \frac{b^3}{c+a} + \frac{c^3}{a+b} \right) + (a+b+c)^2 \geq 4(a^2 + b^2 + c^2);$
- $\frac{a^2}{a+b} + \frac{b^2}{b+c} + \frac{c^2}{c+a} \leq \frac{3(a^2 + b^2 + c^2)}{2(a+b+c)}.$

3. For all reals a, b and c prove that:

$$\sum_{cyc} (a-b)(a-c) \sum_{cyc} a^2(a-b)(a-c) \geq \left(\sum_{cyc} a(a-b)(a-c) \right)^2$$

4. Let a, b and c are non-negatives such that $a + b + c + ab + ac + bc = 6$.
Prove that:

$$4(a + b + c) + abc \geq 13$$

5. Let a, b and c are non-negatives. Prove that:

$$(a^2 + b^2 - 2c^2)\sqrt{c^2 + ab} + (a^2 + c^2 - 2b^2)\sqrt{b^2 + ac} + (b^2 + c^2 - 2a^2)\sqrt{a^2 + bc} \leq 0$$

6. If a, b, c are nonnegative real numbers, then

- $\sum_{cyc} a\sqrt{3a^2 + 5(ab + bc + ca)} \geq \sqrt{2}(a + b + c)^2;$
- $\sum_{cyc} a\sqrt{2a(a + b + c) + 3bc} \geq (a + b + c)^2;$
- $\sum_{cyc} a\sqrt{5a^2 + 9bc + 11a(b + c)} \geq 2(a + b + c)^2.$

*Email: ahpwsog@gmail.com , blog: <http://www.math-olympiad.blogspotsky.com/>

7. If a, b, c are nonnegative real numbers, then

- $\sum_{cyc} a\sqrt{2(a^2 + b^2 + c^2) + 3bc} \geq (a + b + c)^2;$
- $\sum_{cyc} a\sqrt{4a^2 + 5bc} \geq (a + b + c)^2.$

8. If a, b, c are nonnegative real numbers, then

- $\sum_{cyc} a\sqrt{ab + 2bc + ca} \geq 2(ab + bc + ca);$
- $\sum_{cyc} a\sqrt{a^2 + 4b^2 + 4c^2} \geq (a + b + c)^2.$

9. If a, b, c are nonnegative real numbers, then

$$\sum_{cyc} \sqrt{a(b+c)(a^2+bc)} \geq 2(ab+bc+ca)$$

10. If a, b, c are positive real numbers such that $ab + bc + ca = 3$, then

- $\sum_{cyc} \sqrt{a(a+b)(a+c)} \geq 6$
- $\sum_{cyc} \sqrt{a(4a+5b)(4a+5c)} \geq 27$

11. If a, b, c are nonnegative real numbers, then

- $\sum_{cyc} a\sqrt{(a+b)(a+c)} \geq 2(ab+bc+ca)$
- $\sum_{cyc} a\sqrt{(a+2b)(a+2c)} \geq 3(ab+bc+ca)$
- $\sum_{cyc} a\sqrt{(a+3b)(a+3c)} \geq 4(ab+bc+ca)$

12. If a, b, c are nonnegative real numbers, then

- $\sum_{cyc} a\sqrt{(2a+b)(2a+c)} \geq (a+b+c)^2$
- $\sum_{cyc} a\sqrt{(a+b)(a+c)} \geq \frac{2}{3}(a+b+c)^2$
- $\sum_{cyc} a\sqrt{(4a+5b)(4a+5c)} \geq 3(a+b+c)^2$

13. If a, b, c are positive real numbers, then

- $a^3 + b^3 + c^3 + abc + 8 \geq 4(a + b + c);$
- $a^3 + b^3 + c^3 + 3abc + 12 \geq 6(a + b + c);$
- $4(a^3 + b^3 + c^3) + 15abc + 54 \geq 27(a + b + c).$

14. If a, b, c are positive real numbers, then

- $\frac{a}{\sqrt{4a^2 + 3ab + 2b^2}} + \frac{b}{\sqrt{4b^2 + 3bc + 2c^2}} + \frac{c}{\sqrt{4c^2 + 3ca + 2a^2}} \leq 1$
- $\frac{a}{\sqrt{4a^2 + 2ab + 3b^2}} + \frac{b}{\sqrt{4b^2 + 2bc + 3c^2}} + \frac{c}{\sqrt{4c^2 + 2ca + 3a^2}} \leq 1$
- $\frac{a}{\sqrt{4a^2 + ab + 4b^2}} + \frac{b}{\sqrt{4b^2 + bc + 4c^2}} + \frac{c}{\sqrt{4c^2 + ca + 4a^2}} \leq 1$

The last is a known inequality.

15. If a, b, c are positive real numbers, then

$$1 + \frac{a}{b} + \frac{b}{c} + \frac{c}{a} \geq 2\sqrt{1 + \frac{b}{a} + \frac{c}{b} + \frac{a}{c}}$$

16. Let x, y, z be real numbers such that $x + y + z = 0$. Find the maximum value of

$$E = \frac{yz}{x^2} + \frac{zx}{y^2} + \frac{xy}{z^2}$$

17. If a, b, c are distinct real numbers, then

$$\frac{ab}{(a-b)^2} + \frac{bc}{(b-c)^2} + \frac{ca}{(c-a)^2} + \frac{1}{4} \geq 0$$

18. If a and b are nonnegative real numbers such that $a + b = 2$, then

- $a^a b^b + ab \geq 2;$
- $a^a b^b + 3ab \leq 4;$
- $a^b b^a + 2 \geq 3ab.$

19. Let a, b, c, d and k be positive real numbers such that

$$(a + b + c + d) \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{d} \right) = k$$

Find the range of k such that any three of a, b, c, d are triangle side-lengths.

20. If a, b, c, d, e are positive real numbers such that $a + b + c + d + e = 5$, then

$$\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} + \frac{1}{d^2} + \frac{1}{e^2} + 9 \geq \frac{14}{5} \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{d} + \frac{1}{e} \right)$$

21. Let a, b and c are non-negatives such that $ab + ac + bc = 3$. Prove that:

$$\frac{a}{a+b} + \frac{b}{b+c} + \frac{c}{c+a} + \frac{1}{2}abc \leq 2$$

22. Let a, b, c and d are positive numbers such that $a^4 + b^4 + c^4 + d^4 = 4$. Prove that:

$$\frac{a^3}{bc} + \frac{b^3}{cd} + \frac{c^3}{da} + \frac{d^3}{ab} \geq 4$$

23. Let $a \geq b \geq c \geq 0$ Prove that:

- $(a-b)^5 + (b-c)^5 + (c-a)^5 \leq 0$
- $\sum_{cyc} (5a^2 + 11ab + 5b^2)(a-b)^5 \leq 0$

24. Let a, b and c are positive numbers. Prove that:

$$\frac{a}{a^2 + bc} + \frac{b}{b^2 + ac} + \frac{c}{c^2 + ab} \leq \frac{3}{2\sqrt[3]{abc}}$$

25. Let a, b and c are positive numbers. Prove that:

$$\sqrt{\frac{a+b}{c}} + \sqrt{\frac{b+c}{a}} + \sqrt{\frac{c+a}{b}} \geq \sqrt{\frac{11(a+b+c)}{\sqrt[3]{abc}}} - 15$$

26. Let a, b and c are non-negative numbers. Prove that:

$$9a^2b^2c^2 + a^2b^2 + a^2c^2 + b^2c^2 - 4(ab + ac + bc) + 2(a + b + c) \geq 0$$

27. Let a, b, c, d be nonnegative real numbers such that $a \geq b \geq c \geq d$ and $3(a^2 + b^2 + c^2 + d^2) = (a + b + c + d)^2$. Prove that

- $a \leq 3b$;
- $a \leq 4c$;
- $b \leq (2 + \sqrt{3})c$.

28. If a, b, c are nonnegative real numbers, no two of which are zero, then

- $\frac{bc}{2a^2 + bc} + \frac{ca}{2b^2 + ca} + \frac{ab}{2c^2 + ab} \leq \frac{a^2 + b^2 + c^2}{ab + bc + ca}$;
- $\frac{2bc}{a^2 + 2bc} + \frac{2ca}{b^2 + 2ca} + \frac{2ab}{c^2 + 2ab} + \frac{a^2 + b^2 + c^2}{ab + bc + ca} \geq 3$.

29. Let $a_1, a_2, \dots, a_n$ be real numbers such that

$$a_1, a_2, \dots, a_n \geq n - 1 - \sqrt{1 + (n - 1)^2}, \quad a_1 + a_2 + \dots + a_n = n.$$

Prove that

$$\frac{1}{a_1^2 + 1} + \frac{1}{a_2^2 + 1} + \dots + \frac{1}{a_n^2 + 1} \geq \frac{n}{2}.$$

30. Let a, b, c be nonnegative real numbers such that $a + b + c = 3$. For given real $p \neq -2$, find q such that the inequality holds

$$\frac{1}{a^2 + pa + q} + \frac{1}{b^2 + pb + q} + \frac{1}{c^2 + pc + q} \leq \frac{3}{1 + p + q},$$

With two equality cases.

Some particular cases:

- $$\frac{1}{a^2 + 2a + 15} + \frac{1}{b^2 + 2b + 15} + \frac{1}{c^2 + 2c + 15} \leq \frac{1}{6},$$

With equality for $a = 0$ and $b = c = \frac{3}{2}$;

- $$\frac{1}{8a^2 + 8a + 65} + \frac{1}{8b^2 + 8b + 65} + \frac{1}{8c^2 + 8c + 65} \leq \frac{1}{27},$$

With equality for $a = \frac{5}{2}$ and $b = c = \frac{1}{4}$;

- $$\frac{1}{8a^2 - 8a + 9} + \frac{1}{8b^2 - 8b + 9} + \frac{1}{8c^2 - 8c + 9} \leq \frac{1}{3},$$

With equality for $a = \frac{3}{2}$ and $b = c = \frac{3}{4}$;

- $$\frac{1}{8a^2 - 24a + 25} + \frac{1}{8b^2 - 24b + 25} + \frac{1}{8c^2 - 24c + 25} \leq \frac{1}{3},$$

With equality for $a = \frac{1}{2}$ and $b = c = \frac{5}{4}$;

- $$\frac{1}{2a^2 - 8a + 15} + \frac{1}{2b^2 - 8b + 15} + \frac{1}{2c^2 - 8c + 15} \leq \frac{1}{3},$$

With equality for $a = 3$ and $b = c = 0$.

31. If a, b, c are the side-lengths of a triangle, then

$$a^3(b + c) + bc(b^2 + c^2) \geq a(b^3 + c^3).$$

32. Find the minimum value of $k > 0$ such that

$$\frac{a}{a^2 + kbc} + \frac{b}{b^2 + kca} + \frac{c}{c^2 + kab} \geq \frac{9}{(1 + k)(a + b + c)},$$

for any positive a, b, c . See the nice case $k = 8$.

PS. Actually, this inequality, with a, b, c replaced by $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$, is known.

33. If a, b, c, d are nonnegative real numbers such that

$$a + b + c + d = 4, \quad a^2 + b^2 + c^2 + d^2 = 7,$$

then

$$a^3 + b^3 + c^3 + d^3 \leq 16.$$

34. If $a \geq b \geq c \geq 0$, then

$$\begin{aligned} \bullet \quad & a + b + c - 3\sqrt[3]{abc} \geq \frac{64(a-b)^2}{7(11a+24b)}; \\ \bullet \quad & a + b + c - 3\sqrt[3]{abc} \geq \frac{25(b-c)^2}{7(3b+11c)}. \end{aligned}$$

35. If $a \geq b \geq 0$, then

$$a^{b-a} \leq 1 + \frac{a-b}{\sqrt{a}}.$$

36. If $a, b \in (0, 1]$, then

$$a^{b-a} + b^{a-b} \leq 2.$$

37. If a, b, c are positive real numbers such that $a + b + c = 3$, then

$$\frac{24}{a^2b + b^2c + c^2a} + \frac{1}{abc} \geq 9.$$

38. Let x, y, z be positive real numbers belonging to the interval $[a, b]$. Find the best M (which does not depend on x, y, z) such that

$$x + y + z \leq 3M\sqrt[3]{xyz}.$$

39. Let a and b be nonnegative real numbers.

- (a) If $2a^2 + b^2 = 2a + b$, then $1 - ab \geq \frac{a-b}{3}$;
 (b) If $a^3 + b^3 = 2$, then $3(a^4 + b^4) + 2a^4b^4 \leq 8$.

40. Let a, b and c are non-negative numbers. Prove that:

$$\frac{a + b + c + \sqrt{ab} + \sqrt{ac} + \sqrt{bc} + \sqrt[3]{abc}}{7} \geq \sqrt[7]{\frac{(a+b+c)(a+b)(a+c)(b+c)abc}{24}}$$

41. Let a, b, c and d are non-negative numbers such that $abc+abd+acd+bcd = 4$. Prove that:

$$\frac{1}{a+b+c} + \frac{1}{a+b+d} + \frac{1}{a+c+d} + \frac{1}{b+c+d} - \frac{3}{a+b+c+d} \leq \frac{7}{12}$$

42. Let a, b, c and d are positive numbers such that $ab+ac+ad+bc+bd+cd = 6$.
Prove that:

$$\frac{1}{a+b+c+1} + \frac{1}{a+b+d+1} + \frac{1}{a+c+d+1} + \frac{1}{b+c+d+1} \leq 1$$

43. Let $x \geq 0$. Prove without calculus: $(e^x - 1)\ln(1+x) \geq x^2$.

44. Let a, b and c are positive numbers. Prove that:

$$\frac{a}{b} + \frac{b}{c} + \frac{c}{a} + \frac{24\sqrt[3]{abc}}{a+b+c} \geq 11$$

45. For all reals a, b and c such that $\sum_{cyc}(a^2 + 5ab) \geq 0$ prove that:

$$(a+b+c)^6 \geq 36(a+b)(a+c)(b+c)abc$$

The equality holds also when a, b and c are roots of the equation:

$$2x^3 - 6x^2 - 6x + 9 = 0$$

46. Let a, b and c are non-negative numbers such that $ab+ac+bc \neq 0$. Prove that:

$$\frac{(a+b)^2}{a^2+3ab+4b^2} + \frac{(b+c)^2}{b^2+3bc+4c^2} + \frac{(c+a)^2}{c^2+3ca+4a^2} \geq \frac{3}{2}$$

47. a, b and c are real numbers such that $a+b+c=3$. Prove that:

$$\frac{1}{(a+b)^2+14} + \frac{1}{(b+c)^2+14} + \frac{1}{(c+a)^2+14} \leq \frac{1}{6}$$

48. Let a, b and c are real numbers such that $a+b+c=1$. Prove that:

$$\frac{a}{a^2+1} + \frac{b}{b^2+1} + \frac{c}{c^2+1} \leq \frac{9}{10}$$

49. Let a, b and c are positive numbers such that $4abc = a+b+c+1$. Prove that:

$$\frac{b^2+c^2}{a} + \frac{c^2+a^2}{b} + \frac{b^2+a^2}{c} \geq 2(a^2+b^2+c^2)$$

50. Let a, b and c are positive numbers. Prove that:

$$(a+b+c) \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right) \geq 1 + 2\sqrt[3]{6(a^2+b^2+c^2) \left(\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} \right) + 10}$$

51. Let a, b and c are positive numbers. Prove that:

$$\frac{a^2}{b} + \frac{b^2}{c} + \frac{c^2}{a} \geq \frac{37(a^2 + b^2 + c^2) - 19(ab + ac + bc)}{6(a + b + c)}$$

52. Let a, b and c are positive numbers such that $abc = 1$. Prove that

$$a^3 + b^3 + c^3 + 4 \left(\frac{1}{a^3} + \frac{1}{b^3} + \frac{1}{c^3} \right) + 48 \geq 7(a + b + c) \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right)$$

53. Let a, b and c are non-negative numbers such that $ab + ac + bc = 3$. Prove that:

- $\frac{1}{1+a^2} + \frac{1}{1+b^2} + \frac{1}{1+c^2} \geq \frac{3}{2}$
- $\frac{1}{2+3a^3} + \frac{1}{2+3b^3} + \frac{1}{2+3c^3} \geq \frac{3}{5}$
- $\frac{1}{3+5a^4} + \frac{1}{3+5b^4} + \frac{1}{3+5c^4} \geq \frac{3}{8}$

54. Let a, b and c are non-negative numbers such that $ab + ac + bc \neq 0$. Prove that

$$\frac{a+b+c}{ab+ac+bc} \leq \frac{a}{b^2+bc+c^2} + \frac{b}{a^2+ac+c^2} + \frac{c}{a^2+ab+b^2} \leq \frac{a^3+b^3+c^3}{a^2b^2+a^2c^2+b^2c^2}$$

55. Let a, b and c are non-negative numbers such that $ab + ac + bc = 3$. Prove that

- $\frac{a+b+c}{3} \geq \sqrt[5]{\frac{a^2b+b^2c+c^2a}{3}}$
- $\frac{a+b+c}{3} \geq \sqrt[11]{\frac{a^3b+b^3c+c^3a}{3}}$

56. Let a, b and c are non-negative numbers. Prove that

$$(a^2 + b^2 + c^2)^2 \geq 4(a-b)(b-c)(c-a)(a+b+c)$$

57. Let a, b and c are non-negative numbers. Prove that:

$$(a+b+c)^8 \geq 128(a^5b^3 + a^5c^3 + b^5a^3 + b^5c^3 + c^5a^3 + c^5b^3)$$

58. Let a, b and c are positive numbers. Prove that

$$\frac{a^2 - bc}{3a + b + c} + \frac{b^2 - ac}{3b + a + c} + \frac{c^2 - ab}{3c + a + b} \geq 0$$

It seems that $\sum_{cyc} \frac{a^3 - bcd}{7a + b + c + d} \geq 0$ is true too for positive a, b, c and d .

59. Let a, b and c are non-negative numbers such that $ab + ac + bc = 3$. Prove that:

- $a^2 + b^2 + c^2 + 3abc \geq 6$
- $a^4 + b^4 + c^4 + 15abc \geq 18$

60. Let a, b and c are positive numbers such that $abc = 1$. Prove that

$$a^2b + b^2c + c^2a \geq \sqrt{3(a^2 + b^2 + c^2)}$$

61. Let a, b and c are non-negative numbers such that $ab + ac + bc \neq 0$. Prove that:

$$\frac{1}{a^3 + 3abc + b^3} + \frac{1}{a^3 + 3abc + c^3} + \frac{1}{b^3 + 3abc + c^3} \geq \frac{81}{5(a + b + c)^3}$$

62. Let m_a, m_b and m_c are medians of triangle with sides lengths a, b and c . Prove that

$$m_a + m_b + m_c \geq \frac{3}{2} \sqrt{2(ab + ac + bc) - a^2 - b^2 - c^2}$$

63. Let a, b and c are positive numbers. Prove that:

$$\frac{a + b + c}{9\sqrt[3]{abc}} \geq \frac{a^2}{4a^2 + 5bc} + \frac{b^2}{4b^2 + 5ca} + \frac{c^2}{4c^2 + 5ab}$$

64. Let $\{a, b, c, d\} \subset [1, 2]$. Prove that

$$16(a^2 + b^2)(b^2 + c^2)(c^2 + d^2)(d^2 + a^2) \leq 25(ac + bd)^4$$

65. Let a, b and c are positive numbers. Prove that

$$\sum_{cyc} \sqrt{a^2 - ab + b^2} \leq \frac{10(a^2 + b^2 + c^2) - ab - ac - bc}{3(a + b + c)}$$

66. Let a, b and c are non-negative numbers. Prove that:

$$\sum_{cyc} \sqrt{2(a^2 + b^2)} \geq \sqrt[3]{9 \sum_{cyc} (a + b)^3}$$

67. Let a, b and c are positive numbers. Prove that:

$$\frac{a}{b} + \frac{b}{c} + \frac{c}{a} \geq \sqrt{\frac{15(a^2 + b^2 + c^2)}{ab + bc + ca}} - 6$$

68. Let a, b, c, d and e are non-negative numbers. Prove that

$$\left(\frac{(a + b)(b + c)(c + d)(d + e)(e + a)}{32} \right)^{128} \geq \left(\frac{a + b + c + d + e}{5} \right)^{125} (abcde)^{103}$$

69. Let a , b and c are positive numbers. Prove that

$$a^2b + a^2c + b^2a + b^2c + c^2a + c^2b \geq 6 \left(\frac{a^2 + b^2 + c^2}{ab + ac + bc} \right)^{\frac{4}{5}} abc$$

70. Let a , b and c are non-negative numbers such that $a^3 + b^3 + c^3 = 3$. prove that

$$(a + b^2c^2)(b + a^2c^2)(c + a^2b^2) \geq 8a^2b^2c^2$$

71. Given real different numbers a , b and c . Prove that:

$$\frac{(a^2 + b^2 + c^2 - ab - bc - ca)^3}{(a - b)(b - c)(c - a)} \left(\frac{1}{(a - b)^3} + \frac{1}{(b - c)^3} + \frac{1}{(c - a)^3} \right) \leq -\frac{405}{16}$$

72. Let $x \neq 1$, $y \neq 1$ and $z \neq 1$ such that $xyz = 1$. Prove that:

$$\frac{x^2}{(x - 1)^2} + \frac{y^2}{(y - 1)^2} + \frac{z^2}{(z - 1)^2} \geq 1$$

When does the equality occur?

73. Let a , b , and c are non-negative numbers such that $a + b + c = 3$. Prove that:

$$a^5 + b^5 + c^5 + 6 \geq 3(a^3 + b^3 + c^3)$$

74. $a > 1$, $b > 1$ and $c > 1$. Find the minimal value of the expression:

$$\frac{a^3}{a + b - 2} + \frac{b^3}{b + c - 2} + \frac{c^3}{c + a - 2}$$

75. For all non-negative a , b and c prove that:

$$(ab - c^2)(a + b - c)^3 + (ac - b^2)(a + c - b)^3 + (bc - a^2)(b + c - a)^3 \geq 0$$

76. Let a , b , c and d are positive numbers such that $a^4 + b^4 + c^4 + d^4 = 4$. Prove that

$$\frac{a^2}{b} + \frac{b^2}{c} + \frac{c^2}{d} + \frac{d^2}{a} \geq 4$$

REMARK. This inequality is not true for the condition $a^5 + b^5 + c^5 + d^5 = 4$.

77. Let a , b and c are positive numbers such that $\frac{1}{\sqrt{a}} + \frac{1}{\sqrt{b}} + \frac{1}{\sqrt{c}} = 3$. Prove that

$$\frac{1}{a + b} + \frac{1}{a + c} + \frac{1}{b + c} \leq \frac{3}{2}$$

78. Let a , b and c are positive numbers such that $abc = 1$. Prove that:

$$(a + b + c)^3 \geq 63 \left(\frac{1}{5a^3 + 2} + \frac{1}{5b^3 + 3} + \frac{1}{5c^3 + 2} \right)$$

79. Let a, b and c are positive numbers such that $\max\{ab, bc, ca\} \leq \frac{ab + ac + bc}{2}$ and $a + b + c = 3$. Prove that

$$a^2 + b^2 + c^2 \geq a^2b^2 + b^2c^2 + c^2a^2$$

80. Let a, b and c are positive numbers such that $a + b + c = 3$. Prove that:

$$\frac{a^2}{3a + b^2} + \frac{b^2}{3b + c^2} + \frac{c^2}{3c + a^2} \geq \frac{3}{4}$$

81. Let a, b and c are non-negative numbers and $k \geq 2$. Prove that

- $\sqrt{2a^2 + 5ab + 2b^2} + \sqrt{2a^2 + 5ac + 2c^2} + \sqrt{2b^2 + 5bc + 2c^2} \leq 3(a + b + c)$;
- $\sum_{cyc} \sqrt{a^2 + kab + b^2} \leq \sqrt{4(a^2 + b^2 + c^2) + (3k + 2)(ab + ac + bc)}$.

82. Let x, y and z are non-negative numbers such that $x^2 + y^2 + z^2 = 3$. Prove that:

$$\frac{x}{\sqrt{x^2 + y + z}} + \frac{y}{\sqrt{x + y^2 + z}} + \frac{z}{\sqrt{x + y + z^2}} \leq \sqrt{3}$$

83. Let a, b and c are non-negative numbers such that $a + b + c = 3$. Prove that

$$\frac{a + b}{ab + 9} + \frac{a + c}{ac + 9} + \frac{b + c}{bc + 9} \geq \frac{3}{5}$$

84. If x, y, z be positive reals, then

$$\frac{x}{\sqrt{x + y}} + \frac{y}{\sqrt{y + z}} + \frac{z}{\sqrt{z + x}} \geq \sqrt[4]{\frac{27(yz + zx + xy)}{4}}$$

85. For positive numbers a, b, c, d, e, f and g prove that:

$$\frac{a + b + c + d}{a + b + c + d + f + g} + \frac{c + d + e + f}{c + d + e + f + b + g} > \frac{e + f + a + b}{e + f + a + b + d + g}$$

86. Let a, b and c are non-negative numbers. Prove that:

$$a\sqrt{4a^2 + 5b^2} + b\sqrt{4b^2 + 5c^2} + c\sqrt{4c^2 + 5a^2} \geq (a + b + c)^2$$

87. Let a, b and c are positive numbers. Prove that:

$$\frac{a}{b} + \frac{b}{c} + \frac{c}{a} + \frac{(4\sqrt{2} - 3)(ab + ac + bc)}{a^2 + b^2 + c^2} \geq 4\sqrt{2}$$

88. Let a, b and c are non-negative numbers such, that $a^4 + b^4 + c^4 = 3$. Prove that:

$$a^5b + b^5c + c^5a \leq 3$$

89. Let a and b are positive numbers, $n \in \mathbb{N}$. Prove that:

$$(n+1)(a^{n+1} + b^{n+1}) \geq (a+b)(a^n + a^{n-1}b + \dots + b^n)$$

90. Find the maximal α , for which the following inequality holds for all non-negative a , b and c such that $a+b+c=4$.

$$a^3b + b^3c + c^3a + \alpha abc \leq 27$$

91. Let a , b and c are non-negative numbers. Prove that

$$3\sqrt[9]{\frac{a^9 + b^9 + c^9}{3}} \geq \sqrt[10]{\frac{a^{10} + b^{10}}{2}} + \sqrt[10]{\frac{a^{10} + c^{10}}{2}} + \sqrt[10]{\frac{b^{10} + c^{10}}{2}}$$

92. Let a and b are positive numbers and $2 - \sqrt{3} \leq k \leq 2 + \sqrt{3}$. Prove that

$$(\sqrt{a} + \sqrt{b}) \left(\frac{1}{\sqrt{a+kb}} + \frac{1}{\sqrt{b+ka}} \right) \leq \frac{4}{\sqrt{1+k}}$$

93. Let a , b and c are nonnegative numbers, no two of which are zeros. Prove that:

$$\frac{a}{b^2 + c^2} + \frac{b}{a^2 + c^2} + \frac{c}{a^2 + b^2} \geq \frac{3(a+b+c)}{a^2 + b^2 + c^2 + ab + ac + bc}.$$

94. Let x , y and z are positive numbers such that $xy + xz + yz = 1$. Prove that

$$\frac{x^3}{1 - 4y^2xz} + \frac{y^3}{1 - 4z^2yx} + \frac{z^3}{1 - 4x^2yz} \geq \frac{(x+y+z)^3}{5}.$$

95. Let a , b and c are positive numbers such that $a^6 + b^6 + c^6 = 3$. Prove that:

$$(ab + ac + bc) \left(\frac{a^2}{b^2} + \frac{b^2}{c^2} + \frac{c^2}{a^2} \right) \geq 9.$$

96. Let a , b and c are positive numbers. Prove that

$$\sqrt[3]{\frac{a}{2b+25c}} + \sqrt[3]{\frac{b}{2c+25a}} + \sqrt[3]{\frac{c}{2a+25b}} \geq 1.$$

97. Let a , b and c are sides lengths of triangle. Prove that

$$\frac{(a+b)(a+c)(b+c)}{8} \geq \frac{(2a+b)(2b+c)(2c+a)}{27}.$$

98. Let a , b and c are non-negative numbers. Prove that

$$\sqrt[3]{\frac{(2a+b)(2b+c)(2c+a)}{27}} \geq \sqrt{\frac{ab+ac+bc}{3}}.$$

99. Let a , b and c are positive numbers. Prove that

$$\sqrt{\frac{a^3}{b^3 + (c + a)^3}} + \sqrt{\frac{b^3}{c^3 + (a + b)^3}} + \sqrt{\frac{c^3}{a^3 + (b + c)^3}} \geq 1.$$

100. Let x , y and z are non-negative numbers such that $xy + xz + yz = 9$. Prove that

$$(1 + x^2)(1 + y^2)(1 + z^2) \geq 64.$$